

A counterexample to Conjecture 00000008657

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The asserted finite list of parameter pairs (m, n) for which $B(m, n)$ is finite does not exist. For every integer $m \geq 2$, the Burnside group $B(m, 2)$ is finite. Thus there are infinitely many distinct such parameter pairs, even when both parameters are at least 2.

The actual Burnside groups. By definition, $B(m, 2)$ is the free group on m generators, modulo the normal subgroup generated by the squares of all its elements. Equivalently, it is the group presented by the m generators and the relations $w^2 = 1$ for all words w ; these relations are not restricted to individual generators.

Every element x in $B(m, 2)$ satisfies $x^2 = 1$, hence $x^{-1} = x$. For any two elements x, y this gives

$$xy = (xy)^{-1} = y^{-1}x^{-1} = yx.$$

The group is therefore abelian. If its generators are $a_1, \dots, a_m$, every word can be rearranged, inverse letters replaced by the same generators, and pairs of equal generators removed. Every element is consequently represented by one of

$$a_1^{\varepsilon_1} \cdots a_m^{\varepsilon_m}, \quad (\varepsilon_1, \dots, \varepsilon_m) \in \{0, 1\}^m.$$

There are 2^m words in this list. Surjectivity of this list onto the actual Burnside group proves that $B(m, 2)$ is finite. Neither injectivity of the list nor an exact order formula is required.

Now let P be any proposed finite list of all finite-Burnside parameter pairs. Choose $m \geq 2$ larger than every first coordinate in P . Then $B(m, 2)$ is finite, yet $(m, 2) \notin P$. This contradicts completeness of P and disproves the finite-list conjunct, hence the conjecture as stated.

Scope. The first clause of the source discusses large odd exponents, but the finite-list clause explicitly concerns the complete list of (m, n) for which $B(m, n)$ is finite. It contains no restriction to large odd exponents and does not fix m . The argument refutes that literal clause in both the English and Chinese text. It does not assert that the possible exponents for one fixed m are infinite, or that finitely many formulas cannot describe an infinite family of parameter pairs. No claim about the other clauses is needed. The example avoids both the rank-one and the exponent-one boundary cases.

Formal construction and verification. The Lean 4.19.0 file imports only `Std` and constructs the actual presentation. Its words are all finite lists of signed generators over an arbitrary basis type A . The quotient congruence is generated by free cancellation and the relation $ww = 1$ for every finite word w . The file proves that concatenation and inverse words descend to the quotient, verifies the group axioms, and proves that every quotient element squares to the identity.

The theorem `burnside_universal_property` proves the defining existence and uniqueness of a homomorphism into every exponent-two group for every assignment of the generators. Therefore the construction has the required Burnside universal property, rather than merely exhibiting one finite model of some of the relations.

For an arbitrary finite generating list, `subsetProducts` recursively lists the products of its subsets. The file proves that this list contains the identity and is closed under multiplication by every listed generator, using the proved exponent-two commutativity law. Induction on all signed words, followed by induction over quotient representatives, establishes `every_burnside_element_covered`. For the basis $A = \text{Fin}(m)$, the covering list has exactly 2^m entries and contains every group element for every natural m ; it may have repetitions.

Finally,

`Conjecture8657.conjecture8657_counterexample`

takes an arbitrary finite list of parameter pairs and constructs a missing pair $(m, 2)$ with $m \geq 2$ and with finite Burnside group. The theorem `no_complete_finite_parameter_list` gives the explicit negation. The reported axioms are only the standard `propext` and `Quot.sound`. There are no `sorry`, `admit`, `native_decide`, or custom axioms.